JANANI PARAMESWARAN

AI Engineer | 3.5 Years Building LLM, RAG and Multi-Agent Systems on Google Cloud

+91 63793 67941 · jananiparameswaranp@gmail.com · https://www.linkedin.com/in/janani-parameswaran-25a98a2aa
Trichy, Tamil Nadu, India · Remote · https://github.com/Janani-Balakrishnan

SUMMARY

AI Engineer with 3.5 years building LLM applications in Python: RAG pipelines, multi-agent systems, and ML-driven automation on Google Cloud. Works across the full stack, from agent orchestration, retrieval, and evaluation to React frontends, payments, and CI/CD.

Built an AI-powered e-learning platform from zero to one as its sole developer, with multi-agent pipelines that plan curricula, write lessons, and generate course videos. Also built an LLM operations chatbot for a fleet management SaaS covering 1,500+ vehicles across 33 clients, now being extended into intelligent transport planning.

EXPERIENCE

American Online Technological Systems — AI Engineer

Remote · Mar 2023 – Present

Build LLM, RAG, and multi-agent products end to end on two platforms: an AI-powered e-learning platform built solo from zero to one, and an AI-enabled fleet management SaaS serving 33 clients.

AI-Powered E-Learning Platform · *Zero-to-one build, sole developer*

- **RAG content engine:** Built the AI core in Python (FastAPI, PostgreSQL), generating context-aware learning content with LLMs and RAG.
- **Multi-agent pipelines:** Orchestrated agents for curriculum planning, lesson content, and AI course-video generation using Gemini, Veo, and Google TTS/STT.
- **Legal-research assistant:** Built an in-platform assistant using LLMs and RAG grounded in legal documents.
- **Hallucination controls:** Added retrieval relevance checks, source-cited answers, and an explicit "not found" fallback when context is insufficient.
- **Agent reliability:** Validated structured outputs between agents with retries, so a malformed step fails safely instead of corrupting downstream lessons or videos.
- **University data pipelines:** Built Firecrawl and Selenium pipelines that scrape and structure data from US university websites, powering analytics that help students compare and choose universities.
- **Learner experience:** Built Study Planner, Courses, AI Chat, Notebook, Quiz, and Flashcard modules in React and TypeScript.
- **Cloud and payments:** Deployed Docker-based FastAPI microservices on Google Cloud Run with CI/CD through Cloud Build and GitHub Actions, plus Stripe subscriptions and Google, Microsoft, and Facebook OAuth with JWT.

AI-Enabled Fleet Management SaaS · *1,500+ vehicles, 33 clients*

- **Operations chatbot:** Built an LLM-powered assistant that lets fleet owners and CEOs query live data on trips, vehicles, and drivers in natural language, by text or voice, across 1,500+ vehicles for 33 clients.
- **Grounded, tenant-safe answers:** Routed every answer through tenant-scoped, read-only queries on the platform's records, so figures come from the data rather than the model and no client can see another's fleet.
- **Intelligent Transport Planning:** Extending the chatbot from reporting on operations to helping planners plan trips and allocate vehicles and drivers.
- **Order automation:** For a dairy-industry client, orders arriving on Telegram are parsed by an LLM, fuzzy-matched to customers and products, and posted straight into the system, cutting daily manual entry from 8 hours to 4.

AI-Assisted Engineering

- Moved development to an AI-assisted workflow with Claude, using it to build features, write and run tests, and review code, with human review and sign-off on every change.

Career Break — Maternity and Childcare

Jan 2021 – Feb 2023

- Returned to full-time work in March 2023 as an AI Engineer.

Mamakalpa Healthcare Pvt. Ltd. — **Software Developer**

Remote · Aug 2020 – Jan 2021 · Company closed during COVID-19

- Built an AI layer for demand forecasting, inventory planning, and reporting on a .NET ERP, rolled out across a chain of 20 medical shops.

Curvelogics Advanced Technology Solutions Pvt. Ltd. — **Data Science Intern**

Trivandrum, Kerala · Aug 2019 – Mar 2020

- Developed an anomaly detection model for household water consumption for a US-based client.

Career Break — **Government Services Exam Preparation**

2016 – 2019

- Prepared full-time for competitive government service examinations after graduation.

SKILLS

Generative AI	Generative AI, LLMs, LLM APIs (Gemini), Prompt Engineering, Structured Outputs, Guardrails
RAG & Retrieval	RAG, Embeddings, Semantic Search, Vector Databases (Chroma, FAISS)
AI Agents	AI Agents, Multi-Agent Workflows, Tool / Function Calling, LangGraph
Machine Learning	Python, Deep Learning, TensorFlow, PyTorch
Frameworks	LangChain, LangGraph, FastAPI, SQLAlchemy, Django, Streamlit, React, TypeScript
LLM Evaluation	LLM Evaluation and Observability with Langfuse
Databases	PostgreSQL, MongoDB, Chroma, FAISS
Google Cloud	Vertex AI, Gemini, Cloud Run, Cloud SQL, Cloud Storage
Speech AI	Google TTS, Google STT
DevOps	Docker, Git, GitHub Actions, Cloud Build, pytest
APIs & Integrations	REST, Webhooks, Stripe, OAuth, JWT
AI-Assisted Development	Claude for code generation, test writing, and code review

EDUCATION & CERTIFICATION

B.E., Electronics and Communication Engineering Sri Sairam Institute of Technology, Chennai · 2016

- Building Ambient Agents with LangGraph, LangChain Academy · 2025
- Generative AI Foundations, upGrad · 2024
- Applied Data Science with Python Specialization, University of Michigan · 2020
- Deep Learning, Data Science Academy · 2020
- Data Science with Python, Data Science Academy · 2019